EPSR-test-ex59 fluctuations: no accumulation, seeded random numbers

Program(s): fluctuations | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the no-accumulation branch of `fluctuations` [1], which samples the number N of molecule centres in spheres of radius `rmax` placed at random positions in the box, the distribution of N - <N>, and the averages <N>, <N²> and S(0) = (<N²> - <N>²)/<N>, the long-wavelength limit of the structure factor, related to the isothermal compressibility [2], and the start of the generator from a saved `randomno.dat` state.

Code exercised

Fortran (original, fortran_src/):

- `fluctuations.f`: `run_fluctuations` (averages, S(0), list-directed real*8 output), `find_fluctuations` (random spheres, counting)
- `auxiliary_routines.f`: `initran1`, `ran1`, `saveran1` (generator state in `randomno.dat`)
- `read_and_save_files.f`: `outgofz`, `save_files_others`; `setup_symbols_generic.f`: `setup_symbols_fluctuations`

Rust (port, rust/epsr-core/src/):

- `tools/fluctuations.rs`: `run`, `find_fluctuations`
- `rng.rs`: `Ran1::ran1`; `tools/anal_common.rs`: `initran1_file`, `saveran1_file`, `outgofz`
- `tools/anal_common.rs`: `set_sv_dble`; `symbols.rs`: `setup_input_file`, `read_input_file`, `save_inp_file`; `output.rs`: `r_int`

Physical system

Analysed configuration: `200DMSO400Water.ato`, an unchanged copy of the EPSR-test box of 200 dimethyl sulphoxide (DMSO; atom types S, O, C, H) and 400 water molecules (O1, H1): 3200 atoms, DMSO mole fraction 1/3, 300 K, cubic box of side 33.1213 Å (0.0881 atoms/Å³), the EPSR model of the deuterated DMSO-water neutron data (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), cf. [3]. One sphere size, radius 7.5 Å, 1500 random centres (Park-Miller generator with Bays-Durham shuffle [4,5]); the reference gives <N> = 29.26 molecules and S(0) = 0.803 for the last pass. For comparison, a uniform fluid of 600 molecules in 36,334 Å³ (0.0165 molecules/Å³) gives 29.2 molecules in such a sphere.

Inputs

`single.FLUCTUATIONS.dat`: `nsize 40`, `nsumt -1`, `nsamples 1500`, `rmax 7.5`, initial `ntotal 20.0` and `nsqtotal 410.0`; `randomno.dat` copied from ex56; `200DMSO400Water.ato` copied unchanged from EPSR-test.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
`repeat 2`. With `nsumt -1` (the nsumts < 0 branch) nothing is accumulated and nsumt stays -1; the given `ntotal`/`nsqtotal` are ignored because nsumt is 0 internally, so the reference `ntotal` (29.26) is the mean of the current pass only. `initran1` reads the saved state, and pass 2 continues from the state saved by pass 1. Compared after each pass: `single.FLUCTUATIONS.n01`, `single.FLUCTUATIONS.dat`, `randomno.dat`.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] J.-P. Hansen and I. R. McDonald, Theory of Simple Liquids, 3rd ed., Academic Press (2006).
[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[4] S. K. Park and K. W. Miller, "Random number generators: good ones are hard to find", Commun. ACM 31, 1192-1201 (1988).
[5] W. H. Press, S. A. Teukolsky, W. T. Vetterling and B. P. Flannery, Numerical Recipes in Fortran 77, 2nd ed., Cambridge University Press (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.